

NOVA CART

Problem Diagnosis, Prompt Journey and Recommendations

The diagnosis in one sentence. NOVA CART is acquiring customers faster than it can keep its promise to them. The promise ("it's in stock, and it will arrive on time") breaks because independent stores' inventory data goes stale. Failed early orders stop customers from reaching the habit threshold of three orders, and rising discounts hide the damage while making growth more expensive.

65% of revenue now goes to promotions (was 44%)

27% repeat purchase rate (was 41%)

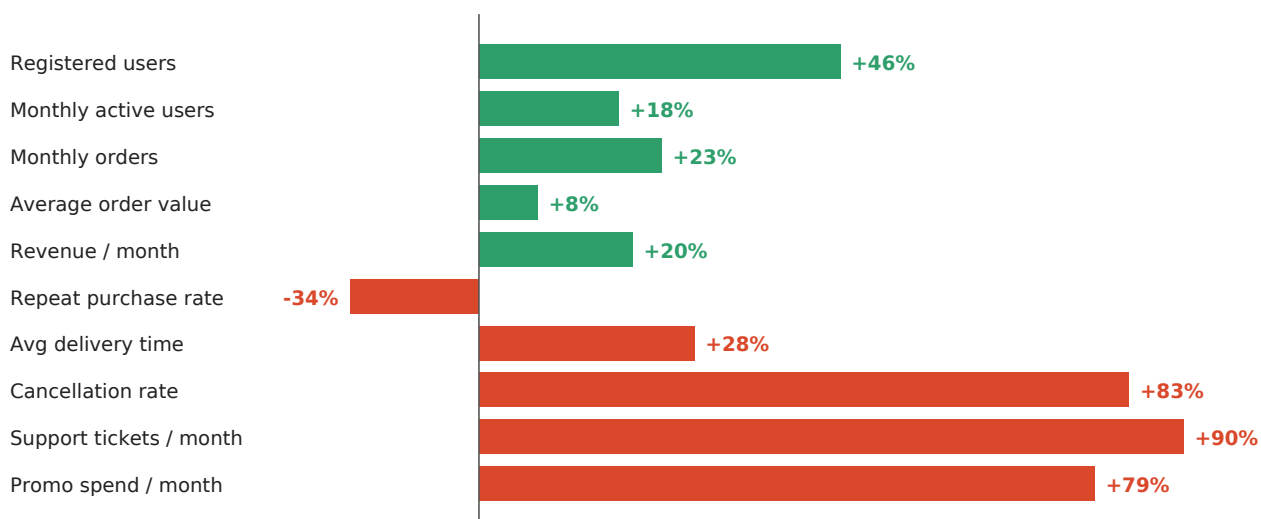
11% of orders cancelled (was 6%), mostly from stock and delay

61% of customers who left had rated NOVA CART 4★ or higher

PART 1 | PROBLEM DIAGNOSIS

1. Growth looks healthy, but quality is falling

Six-month change in each metric. Green bars are metrics where an increase is good; red bars are metrics where the direction is bad for the business. **Every red bar is moving the wrong way, and most are moving faster than the green ones.**



Derived measure (from the case data)	6 months ago	Now	What it shows
Active users as a share of registered users	48%	38%	Sign-ups are not turning into regular users
Promo spend as a share of revenue	44%	65%	Most of every rupee earned is given back
Promo spend per order	₹30	₹44	+45%: each order costs more to win
Support tickets per 1,000 orders	99	153	Experience per order is worse, not just busier
Revenue after promo spend	₹12.3L	₹9.1L	Down about 26% while headline revenue is up 20%

Promo spend rose ₹7.5L while revenue rose ₹4.3L, so each extra ₹1 of revenue cost about ₹1.74 in promotions. That is roughly ₹103 of extra promotion for every additional monthly order.

2. The evidence chain

Finding 1: Customers leave before they build a habit

- Only **54%** of new users complete a first order, and only **31%** place a second order within 30 days.
- Customers who complete **3 orders** have a **72%** chance of ordering again the next month. The business is losing people before the point where retention takes hold.

- **61%** of customers who stopped ordering had rated NOVA CART 4★ or higher. Ratings did not predict leaving, so customers did not dislike the app. Something is stopping them from returning.
- Customers acquired through large first-order discounts retain worse than organic users, and **44%** of coupons are never redeemed, so discounts are an expensive and weak lever.

Finding 2: The promise breaks on availability and delivery time

Of the **4,235** orders cancelled each month (11% of 38,500), the reasons are:

Cancellation reason	Share of cancellations	Approx. orders / month	Likely driver
Product unavailable	35%	1,480	Stale stock data
Customer cancelled because of delay	27%	1,140	Over-optimistic ETA, partner shortage
Store rejected the order	18%	760	Busy stores, no stock confidence
Delivery partner unavailable	12%	510	Partner capacity
Other	8%	340	

- About **53%** of cancellations (unavailable + store rejection) trace back to inventory and store readiness, and about **39%** to delay and partner capacity.
- **13%** of orders arrive more than 15 minutes late, and **8%** contain a substituted item. In the survey, **29%** said products shown as available became unavailable after ordering, and **34%** said delivery takes too long.
- Many partner stores update stock manually, some only every **1-3 days**, so the app shows items that are not on the shelf.

Finding 3: Failed orders flood support, and support is slow

Ticket type	Share	Approx. tickets / month
Refund status	29%	1,710
Delayed delivery	24%	1,420
Missing / unavailable products	19%	1,120
Coupon problems	13%	770
Incorrect orders	9%	530
Other	6%	350

- About **81%** of tickets (refund, delay, missing and incorrect items) are consequences of failed or late orders. Another **13%** come from confusing coupons.
- Average resolution takes **9.2 hours**, and support uses separate systems for orders, refunds and store communication. Slow refunds add a second bad experience on top of the failed order.

Finding 4: Stores are both the bottleneck and the advantage

- From 100 partner interviews: **46%** value the extra customers, but **39%** say maintaining online inventory takes too much effort, **31%** say promotions cut their margins, **28%** cannot predict what will sell online, and **23%** reject orders in busy periods.
- **18%** are considering leaving within a year (roughly 110 of 620 stores if the sample is representative). Losing stores would remove the assets that make NOVA CART different.
- **19%** of customers repeatedly search for products not available nearby, **18%** find it hard to discover local products, and **21%** prefer buying directly from nearby stores. Customers who shop across multiple store categories repeat more.
- Large competitors can beat NOVA CART on speed and discount size. They cannot easily match hundreds of independent local stores. The company has no clear strategy around this yet.

Finding 5: More marketing would make the leak bigger

- About **58%** of the ₹17L budget (≈₹9.9L) goes to acquiring new customers, and management is considering a further **30%** increase.

- A 30% increase is about ₹5.1L more per month, roughly ₹30.6L over six months, which is more than the entire ₹25L implementation budget. It would pay to bring more people into a funnel where most customers do not place a second order within 30 days.

3. Leadership views tested against the evidence

View	Supporting evidence	Verdict
CEO: improve retention	Repeat rate 41% → 27%; only 31% place a second order within 30 days	Right goal , but it is the outcome, not the cause
Marketing: more acquisition	Registered users +46%	Weak : active users +18%, discounted users retain worse, 44% of coupons unused
Operations: delivery reliability	37 min average; 13% late by 15+ min; 27% of cancellations from delay	Partly right : delay explains about 39% of cancellations
Partners: inventory accuracy	35% of cancellations from unavailable products; stock updated every 1–3 days	Strongest single cause : also drives store rejections and missing-item tickets
Product: local commerce advantage	19% search for unavailable local items; multi-category users repeat more	Right and the main opportunity, but only works if listings are trustworthy
Finance: quality and cost of growth	Promo 65% of revenue; revenue after promo down 26%	Right : confirmed by the numbers

All six views are partly correct, and most connect to one chain: **stale inventory → failed or late orders → lost trust and support load → no second or third order → growth bought with discounts.**

4. Alternative explanations considered

- **"Prices and fees are too high."** This is the top survey item (38%), but the wording is that prices feel higher *than expected*, which points to unclear fees and confusing discounts (24%) rather than a need for lower prices. Spending 65% of revenue on promotions shows deeper discounting is not affordable.
- **"We need faster delivery."** Delivery time matters (34%), but partner shortage causes only 12% of cancellations. Competing on pure speed against larger rivals is unlikely to succeed, whereas an honest, reliable ETA is within reach.
- **"We need bigger first-order offers."** First-order discounts already attract customers who retain worse, and the larger drop-off happens after the first order, which a bigger first-order offer does not address.

5. Underlying problem and opportunity

Problem statement: How can NOVA CART make its availability and delivery promise trustworthy, so that new customers reach their third order, while reducing the cost of growth and keeping partner stores effort-light and engaged?

Opportunity: Turn the weakness (unreliable local inventory) into the differentiator: a platform where customers can trust what is available nearby, see honest delivery times, find local products big apps do not carry, and get a better reason to return than a coupon.

Target users: first-time and early-stage customers (orders 1–3), and independent partner stores that maintain the catalogue.

6. Limits of the evidence

- Orders × average order value (38,500 × ₹486) is about ₹18.7L, below the stated ₹26.1L revenue, so revenue likely includes fees, commissions or other income. Revenue-based ratios above use the figures as given.
- The brief does not say whether the 31% second-order figure is measured against all new users or only those who completed a first order, and the survey does not show how answers split between active and churned customers.
- Cancellation counts, ticket counts and the roughly 110 at-risk stores are calculated from percentages and a 100-store sample, so they are estimates.

- Several findings show association, not proven cause. The proposed product should therefore be measured against retention, cancellation rate, ETA accuracy, tickets per 1,000 orders and promo spend as a share of revenue.

PART 2 | PROMPT JOURNEY

These are the key prompts that shaped the analysis and the product, lightly edited for spelling and length. Each shows what the AI surfaced and the decision it changed. The journey also shows a change of mind: the early prompts leaned on offers and discounts, and the later diagnosis showed that discounts are part of the problem, so the product was refined to cap them and focus on reliability.

Stage 1: Turn the scenario into a build prompt

Prompt	<i>"Read this scenario [NOVA CART metrics]. Create a prompt for building NOVA CART, a place to buy groceries, snacks and supermarket items. Include first-order discounts, referral rewards, social media advertising, influencer promotions, festival coupons and free-delivery campaigns; a login page; quick stock updates, demand prediction, popular-product suggestions and low-stock alerts. After login show how many of the 3 free deliveries are left, pop up free delivery for orders of ₹2,000+, let the customer choose a delivery location, and give scratch-card coupons with praise on every ₹500+ purchase. Build repeat usage through reliability, local loyalty rewards, saved neighbourhood shops and better support. Show a realistic delivery time, help customers discover nearby local products, and show an availability confidence label with substitutes before an order fails."</i>
What the AI surfaced	Registered users grew 46% but active users only 18%, and promo spend grew 79% against revenue growth of 20%. Growth looked bought with discounts rather than earned through reliability.
Decision it shaped	The app is built around reliability, local loyalty, saved shops and support, not only discounts. A promo-spend-versus-revenue guardrail was added to the admin dashboard.

Stage 2: Design the new-customer offer

Prompt	<i>"If a customer enters through sign-up (new customer), show that the first 3 orders are free and give 50% off the first order."</i>
What the AI surfaced	An uncapped 50% off would cost ₹1,000 on a ₹2,000 first order, and a separate free-delivery set would give new users 6 free deliveries instead of 3.
Decision it shaped	50% off is capped at ₹150 with a ₹199 minimum cart (admin-configurable). New users share one 3-delivery counter. Scratch-card rewards from the first order work from the second order onward, to encourage a return.

Stage 3: Make delivery honest and visible

Prompt	<i>"Add live tracking of the order, and show the customer 'Receive your order within 10–30 min*', with the * explained as based on the delivery location."</i>
What the AI surfaced	A fixed 10–30 minute promise could repeat the slide from 29 to 37 minutes if real times run longer.
Decision it shaped	The message is location-based, with an honesty rule: if the true estimate is beyond 30 minutes, the app shows the real range. Live tracking includes a moving map marker, an ETA countdown, and a delay credit.

Stage 4: Prepare the build for an AI coding agent

Prompt	<i>"Give the final prompt to build this in Antigravity." Then: "[http://127.0.0.1:5173/] Write a prompt for it and add a login credentials page with smooth animations."</i>
What the AI surfaced	A local app cannot be opened by the AI, so the prompt tells the coding agent to inspect and extend the existing project rather than rebuild it.
Decision it shaped	The build is split into six phases with a plan first and browser-agent testing after each. The login page has mock authentication, animated transitions, validation feedback and reduced-motion support.

Stage 5: Diagnose the problem from the data	
Prompt	"Based on how I made the prompt, give me a Problem Diagnosis: clearly explain the underlying problem or opportunity and the evidence supporting it."
What the AI surfaced	Promo spend rose from 44% to 65% of revenue and from ₹30 to ₹44 per order. Tickets per 1,000 orders rose from 99 to 153, and revenue after promo spend fell about 26%. The AI also flagged that orders × order value (₹18.7L) does not match the stated ₹26.1L revenue.
Decision it shaped	The problem was framed as the quality and cost of growth, not a shortage of users. The revenue mismatch is disclosed as an assumption.
Stage 6: Test the diagnosis against the full case brief	
Prompt	Uploaded the full challenge brief and asked for the Problem Diagnosis as a PDF.
What the AI surfaced	The survey, operations and partner data pointed further upstream: 35% of cancellations are from unavailable products, some stores update stock only every 1–3 days, 61% of churned customers had rated 4★ or higher, and customers who reach 3 orders have a 72% chance of ordering again.
Decision it shaped	The root cause was reframed from "too many discounts" to stale inventory → failed early orders → no third order . Priorities became trustworthy availability, honest delivery times and getting customers to order three.
Stage 7: Check the recommendation to spend more	
Prompt	"Suggest they should increase promotions and bring in more retailers in the city."
What the AI surfaced	Raising total promotion spend is not supported by the evidence (already 65% of revenue, and a 30% increase would add about ₹5.1L a month). Adding retailers helps only if their stock data is trustworthy, since stale catalogues cause the biggest share of cancellations.
Decision it shaped	Recommend increasing promotions that reward repeat orders and local stores, funded by reallocating from broad acquisition, and expanding retailers in waves with easy stock updates (Part 3).

Suggested next prompts (not yet run)

- **Challenge the diagnosis:** "Argue against our diagnosis. What evidence would show that delivery speed or price, not stale inventory, is the main cause?"
- **Size the impact:** "If cancellations from unavailable products fall by half, how many orders, how much revenue and how many support tickets per month does that recover?"
- **Pick where to expand:** "Which product categories and neighbourhoods should we add retailers in first, based on searches for unavailable products and long delivery times?"
- **Reduce store effort:** "Design the shortest possible stock-update flow for a store owner who has 30 seconds between customers."

PART 3 | RECOMMENDATIONS: PROMOTIONS AND RETAILER EXPANSION

Increase promotions that build repeat orders

The evidence supports **more promotion of the right kind**, not simply a larger budget. Raising the total on top of 65% of revenue would repeat the problem, so the increase should be funded by shifting money out of weak spend (58% of the budget goes to new-customer acquisition and 44% of coupons are never redeemed).

Promotion	Recommendation	Why
Welcome offer (50% off, 3 free deliveries)	Keep, capped at ₹150	Wins first orders, but discount-acquired users retain worse, so limit the cost
Rewards for order 2 and order 3	Increase	Customers who reach 3 orders have a 72% chance of ordering again
Scratch cards on ₹500+ orders	Increase , usable from the next order	Rewards larger baskets and brings customers back
Festival coupons	Increase , tied to local stores and regional products	Supports discovery of products big platforms do not carry
Referral rewards	Keep, pay after the friend's first delivered order	Avoids paying for sign-ups that never order

Promotion	Recommendation	Why
Broad coupons, ads and influencer campaigns	Reduce and test in small cohorts	44% of coupons go unused and 58% of spend is on acquisition
Store-funded offers	Share cost with participating stores	31% of stores say promotions reduce their margins

If management still wants to raise total spend, run it as a test in one city or cohort with a comparison group, and keep a ceiling on promo spend as a share of revenue.

Add more retailers in the cities, in a controlled way

- **Where:** neighbourhoods with few stores or long delivery times, and categories customers search for but cannot find (19% repeatedly search for unavailable products): bakery, pharmacy, stationery and regional groceries.
- **Why:** local variety is NOVA CART's difference from larger competitors, and customers who buy across several store categories repeat more. It also needs no warehouses or large hiring.
- **How:** onboard in waves, and make quick tap-based stock updates, bulk upload, low-stock alerts and availability confidence labels part of onboarding, because 39% of current stores say catalogue upkeep takes too much effort.
- **Protect current stores first:** 18% are considering leaving and 23% reject orders in busy periods. Adding stores with stale catalogues would increase the cancellations from unavailable products (35% of the total).
- **Measure:** cancellation rate and stock-update frequency per store, ETA accuracy, second- and third-order rates, share of multi-category customers, and promo spend as a share of revenue.

Suggested order of work: **fix inventory trust and store effort → pilot new retailers in one city → scale to the other cities.**